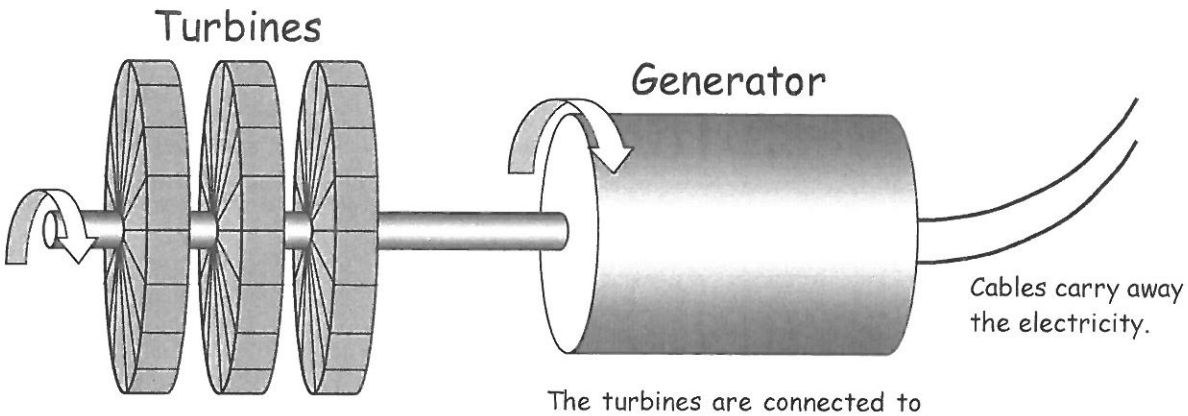
W.S.89. Generating electricity. Name .....

Most people use electricity many times every day. Electricity is GENERATED (produced) in power stations before it is sent to homes and factories. A number of energy resources can be used to generate electricity. Large TURBINES that are connected to GENERATORS are made to turn. As the generators turn they produce the electricity. The diagram below shows how this works.



TURBINES are made to turn by some form of energy. This can be wind, moving water, or steam under high pressure.

The turbines are connected to a GENERATOR. As this spins around it generates electricity.

Many power stations burn coal to heat water. As the water boils steam is produced under high pressure. The turbines are pushed around by the force of the steam. Only about one third of the chemical energy inside the coal is changed into electrical energy. The other two thirds of the energy is lost to the surroundings as heat.
Some power stations use wind power to push the turbines around. In a HYDROELECTRIC power station water rushing downhill is used to turn the turbines.

Exercise - Complete the sentences below.

1) In order to generate electricity turbines must be made to _ _ _ _

2) In many power stations pressure from _ _ _ _ _ is used to turn the turbines.

3) Many power stations use _ _ _ _ as the fuel to heat water.

4) Coal powered electricity stations are wasteful because only about one _ _ _ _ _ of the chemical energy inside the coal is changed into electrical energy.

5) Hydroelectricity is generated by using energy from moving _ _ _ _ _

6) In the future _ _ _ _ and water power may become the main ways of generating electricity because they will never run out.

KS3 Science Revision Worksheets Special Edition © P. Hill, Beaver Educational Resources 2000. Registered to Fitzalan High School. CF11 8XB.